C 23323 (Pages : 3) Name.....

Reg. No.....

FOURTH SEMESTER B.Sc. DEGREE EXAMINATION, APRIL 2017

(CUCBCSS–UG)

Complementary Course

MAT 4C 04—MATHEMATICS

Time : Three Hours Maximum : 80 Marks

Part A

Answer **all** *questions.*
Each question carries 1 *mark.*

1. Find the general solution of $y^{11} + w^2 y = 0$.
2. Find $(D^2 + 3D)\cosh 3x$.
3. Write the general form of the Cauchy-Euler equation.
4. Find $L(a + bt + ct^2)$.
5. $L(f') =$ ———.
   (a) $L(f) - Sf(0)$. (b) $L(f) - f(0)$.
   (c) $SL(f) - f(0)$. (d) $SL(f) - f'(0)$.
6. $L^{-1}\left(\frac{1}{s^2 + a^2}\right) =$ ———.
7. Write the second shifting theorem of Laplace transform.
8. Sketch $f(x) = |x|$ for $-\pi < x < \pi$.
9. Find a_0 in the Fourier series expansion of $f(x) = x^2, -\pi < x < \pi$.
10. Write the Picard's iteration formula to find a numerical solution of $y' = f(x, y), y(x_0) = y_0$.
11. Write the one-dimensional wave equation.
12. Find the period of $\cos \pi x$.

(12 × 1 = 12 marks)

Part B

Answer any **nine** *questions.*
Each question carries 2 *marks.*

13. Find the general solution of $(9D^2 + 6D + 1)y = 0$.
14. Solve $x^2 y'' - 3xy' + 4y = 0$.
15. Find the Laplace transform of $(t + 1)^2 e^t$.

Turn over

16. Find $L^{-1}\left(\frac{60+6s^2+s^4}{s^7}\right)$.

17. If $f(t)=t$ and $g(t)=e^{at}$ find the convolution $(f*g)(t)$.

18. Find the Fourier cosine series of the function $f(x)=\pi-x$ in $0<x<\pi$.

19. Prove that product of an even function and an odd function is an odd function.

20. Apply Picard's method to solve the initial value problem $y'=x^2+y,\ y(0)=-1$.

21. Solve the partial differential equation $u_{xy}=u$.

22. Evaluate $\int_0^6 \frac{dx}{1+x^2}$ using Trapezoidal rule taking $h=1$.

23. Solve the initial value problem $y''+y'=0,\ y(0)=5,\ y'(0)=-3$ using Laplace transform.

24. Derive the Euler's formula to solve the differential equation $y'=f(x,y),\ y(x_0)=y_0$.

(9 × 2 = 18 marks)

Part C

Answer any **six** *questions.*
Each question carries 5 *marks.*

25. Find the general solution of $y''+2y'+y=2x+x^2$.

26. Find the general solution of $(D^2+3D-4)y=8\cos 2x$.

27. Solve $y''+y=\sec x$ by the method of variation of parameters.

28. Find the inverse Laplace transform of $\frac{1}{(s+\sqrt{2})(s-\sqrt{3})}$.

29. Solve the initial value problem by Laplace transform $y'+3y=10\sin t,\ y(0)=0$.

30. Find the Fourier series of the function $f(x)=x^2,\ -\pi<x<\pi$.

31. Using Laplace transform solve the integral equation $y(t)=1-\int_0^t (t-\tau)\,y(\tau)d\tau$.

32. Find the deflection $u(x,t)$ of a string of length $L=2\pi$ when $c^2=1$, the initial velocity is zero and initial deflection is $0.1(\pi^2-x^2)$.

33. Evaluate $\int_1^7 \frac{dx}{x}$ using Simpson's rule by dividing [1, 7] into 6 equal parts.

(6 × 5 = 30 marks)

Part D

Answer any **two** *questions.*
Each question carries 10 *marks.*

34. Solve the initial value problem $y'' + 1.2y' + 0.36y = 4e^{-0.6x}$, $y(0) = 0$, $y'(0) = 1$.

35. Solve $y'' + 2y' + 5y = e^{-t} \sin t$, $y(0) = 0$. Using Laplace transform. Given $y'(0) = 1$.

36. Use improved Euler's method to determine $y(0.2)$ in two steps from $\frac{dy}{dx} = x^2 + y$, given that $y(0) = 1$.

$(2 \times 10 = 20$ marks)